

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110
(An Autonomous Institution, Affiliated to Anna University, Chennai)Department of
Computer Science and Engineering
UCS2603 - Theory of Computation
VI - Sem A, B - Section
Unit – 1, II

1. Construct a DFA and NFA that recognizes the language $C = \{w \in \Sigma^* \mid w = saba \text{ for some string } s \in \Sigma^*\}$

Also check the following words for both NFA & DFA

i) bbabab

ii) baba

2. Construct a DFA and NFA for $E = \{w \in \Sigma^* \mid w \text{ begins with } b \text{ and ends with } a\}$

Also check the following words for both NFA & DFA

i) bbabab

ii) baba

3. Construct the equivalent DFA following NFA given in transition table:

i)

	inputs	
states	0	1
$\rightarrow p$	{p,q}	{p}
q	{r}	{r}
r	{s}	-
*s	{s}	{s}

ii)

	inputs	
states	0	1
$\rightarrow p$	{q,s}	{q}
*q	{r}	{q,r}
r	{s}	{p}
*s	-	{p}

iii)

	inputs	
states	0	1
$\rightarrow *q_0$	{q ₀ ,q ₁ }	{q ₁ }
q ₁	{q ₀ }	{q ₀ ,q ₁ }

4. Construct the equivalent NFA for the ϵ -NFA given in the following transition table:

i)

	inputs		
states	ϵ	a	b
$\rightarrow p$	{q}	{p}	-
q	{r}	-	{q}
*r	-	-	{r}

ii)

	inputs			
states	ϵ	a	b	c
$\rightarrow 0$	{1,2}	{0}	-	
1	{3}	-	{2}	
2	-	-	{3}	
*3	-	-	-	{3}

5. Convert the following Regular Expression to ϵ -NFA.

a. $(a|b)^* abb (a|b)^*$

b. $(a|b)^* a (a|b)$

c. $(a|b|c)^* (a|b)^*$

d. $(a|b) a (a|b)^* abb$

e. $0(2|51)^*$

f. $(ab)^* abc$

6. Convert all ϵ - NFA's in Ex. 5 into DFA.

7. Describe the languages denoted by the following re

a. $0(0|1)^*0$

b. $(0|1)^* 0 (0|1) (0|1)$